ITI 0402 Loogika
Loeng 9

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Predikaatloogika loomulik tuletus

- Predikaatloogika loomulikus tuletuses lisanduvad lauseloogika loomuliku tuletuse reeglitele järgmised:

$$\frac{A[y/x]}{\forall x\ A}\ \forall\mathcal{I} \qquad \frac{\forall x\ A}{A[t/x]}\ \forall\mathcal{E}$$

$$\frac{A[t/x]}{\exists x\ A}\ \exists\mathcal{I} \qquad \frac{\exists x\ A \qquad \begin{array}{c}A[y/x]\\ \vdots \\ C\end{array}}{C}\ \exists\mathcal{E}$$

$\forall\mathcal{I}$ reegli puhul y ei tohi vabalt esineda valemis $\forall x\ A$ ega kasutada olevates hüpoteesides

$\exists\mathcal{E}$ reegli puhul y ei tohi vabalt esineda valemites $\exists x\ A$, C ega kasutada olevates hüpoteesides

Samuti, mõlema reegli puhul y asendused valemisse A ei tohi pärast substitutsiooni osutuda seotuks.

Loomuliku tuletuse näiteid

$$\dfrac{\dfrac{\dfrac{\exists x\ \forall y\ p(x,y)^{+1} \quad \dfrac{\dfrac{\forall y\ p(x',y)^{+2}}{p(x',y')}\,\forall\mathcal{E}}{\exists x\ p(x,y')}\,\exists\mathcal{I}}{\exists x\ p(x,y')}\,\exists\mathcal{E},-2,x' \text{ värske}}{\forall y\ \exists x\ p(x,y)}\,\forall\mathcal{I},y' \text{ värske}}{\exists x\ \forall y\ p(x,y) \supset \forall y\ \exists x\ p(x,y)}\,\supset\mathcal{I},-1$$

$$\dfrac{\dfrac{\dfrac{\exists x\ p(x)^{+1} \quad \dfrac{\dfrac{\forall x\ \neg p(x)^{+2}}{\neg p(x')}\,\forall\mathcal{E} \quad p(x')^{+3}}{\bot}\,\neg\mathcal{E}}{\bot}\,\exists\mathcal{E},-3,x' \text{ värske}}{\neg\forall x\ \neg p(x)}\,\neg\mathcal{I},-2}{\exists x\ p(x) \supset \neg\forall x\ \neg p(x)}\,\supset\mathcal{I},-1$$

Loomuliku tuletuse näiteid

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\neg\forall x\ \neg p(x)}\,{+}1 \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{\neg\exists x\ p(x)}\,{+}2 \qquad \dfrac{}{p(x')}\,{+}4}{\exists x\ p(x)}\,\exists\mathcal{I}}{\bot}\,\neg\mathcal{E}}{\neg p(x')}\,\neg\mathcal{I},-4}{\forall x\ \neg p(x)}\,\forall\mathcal{I}, x' \text{ värske}}{\bot}\,\neg\mathcal{E}}{\exists x\ p(x)}\,\bot\mathcal{E} \qquad \dfrac{}{\exists x\ p(x)}\,{+}3}{\exists x\ p(x)}\ \text{dilemma}, -2, -3}{\neg\forall x\ \neg p(x) \supset \exists x\ p(x)}\,\supset\mathcal{I}, -1$$

Loomuliku tuletuse näiteid

$$\dfrac{\dfrac{\dfrac{\overset{+1}{\exists x\, p(x) \supset q} \quad \dfrac{\overset{+1}{p(x')}}{\exists x\, p(x)}\ \exists\mathcal{I}}{q}\ \supset\mathcal{E}}{p(x') \supset q)}\ \supset\mathcal{I}, -2}{\forall x\,(p(x) \supset q)}\ \forall\mathcal{I}, x' \text{ värske}}{(\exists x\, p(x) \supset q) \supset \forall x\,(p(x) \supset q)}\ \supset\mathcal{I}, -1$$

$$\dfrac{\dfrac{\overset{+1}{\exists x\, p(f(x))} \quad \dfrac{\overset{+2}{p(f(x'))}}{\exists x\, p(x)}\ \exists\mathcal{I}}{\exists x\, p(x)}\ \exists\mathcal{E}, x' \text{ värske}, -2}{(\exists x\, p(f(x)) \supset \exists x\, p(x)}\ \supset\mathcal{I}, -1$$

Predikaatloogika sekventsiarvutus

- Predikaatloogika sekventsiarvutuses lisanduvad lauseloogika sekventsiarvutusele järgmised reeglid kvantorite kohta:

$$\frac{\Gamma \to A[y/x], \Delta}{\Gamma \to \forall x\, A, \Delta}\ \forall\mathcal{R} \qquad \frac{\Gamma, (\forall x\, A,)\, A[t/x] \to \Delta}{\Gamma, \forall x\, A \to \Delta}\ \forall\mathcal{L}$$

$$\frac{\Gamma \to A[t/x], (\exists x\, A,)\, \Delta}{\Gamma \to \exists x\, A, \Delta}\ \exists\mathcal{R} \qquad \frac{\Gamma, A[y/x] \to \Delta}{\Gamma, \exists x\, A \to \Delta}\ \exists\mathcal{L}$$

$\forall\mathcal{R}$ reegli puhul y ei tohi vabalt esineda valemis $\forall x\ A$ ega valemihulkades Γ, Δ.

$\exists\mathcal{L}$ reegli puhul y ei tohi vabalt esineda valemis $\exists x\ A$ ega valemihulkades Γ, Δ.

Samuti, mõlema reegli puhul y asendused valemisse A ei tohi pärast substitutsiooni osutuda seotuks.

Sekventsiarvutuse näiteid

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p(x',y') \to p(x',y')}\,\text{hyp.}}{\forall y\, p(x',y) \to p(x',y')}\,\forall\mathcal{L}}{\forall y\, p(x',y) \to \exists x\, p(x,y')}\,\exists\mathcal{R}}{\exists x \forall y\, p(x,y) \to \exists x\, p(x,y')}\,\exists\mathcal{L}, x' \text{ värske}}{\exists x \forall y\, p(x,y) \to \forall y \exists x\, p(x,y)}\,\forall\mathcal{R}, y' \text{ värske}}{\to \exists x \forall y\, p(x,y) \supset \forall y \exists x\, p(x,y)}\,\supset\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p(x') \to p(x')}\,\text{hyp.}}{p(x'), \neg p(x') \to}\,\neg\mathcal{L}}{p(x'), \forall x \neg p(x) \to}\,\forall\mathcal{L}}{\exists x\, p(x), \forall x \neg p(x) \to}\,\exists\mathcal{L}, x' \text{ värske}}{\exists x\, p(x) \to \neg\forall x \neg p(x)}\,\neg\mathcal{R}}{\to \exists x\, p(x) \supset \neg\forall x \neg p(x)}\,\supset\mathcal{R}$$

Sekventsiarvutuse näiteid

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p(f(x'),y') \to p(f(x'),y')}\,\text{hyp.}}{p(f(x'),y') \to \exists y\ p(f(x'),y)}\,\exists\mathcal{R}}{p(f(x'),y') \to \exists x\ \exists y\ p(x,y)}\,\exists\mathcal{R} \qquad \dfrac{\dfrac{\dfrac{\dfrac{}{p(y',g(x')) \to p(y',g(x'))}\,\text{hyp.}}{p(y',g(x')) \to \exists y\ p(y',y)}\,\exists\mathcal{R}}{p(y',g(x')) \to \exists x\ \exists y\ p(x,y)}\,\exists\mathcal{R}}{p(f(x'),y') \vee p(y',g(x')) \to \exists x\ \exists y\ p(x,y)}\,\vee\mathcal{L}$$

$$\dfrac{\dfrac{\dfrac{p(f(x'),y') \vee p(y',g(x')) \to \exists x\ \exists y\ p(x,y)}{\exists y\ (p(f(x'),y) \vee p(y,g(x'))) \to \exists x\ \exists y\ p(x,y)}\,\exists\mathcal{L}, y' \text{ värske}}{\exists x\ \exists y\ (p(f(x),y) \vee p(y,g(x))) \to \exists x\ \exists y\ p(x,y)}\,\exists\mathcal{L}, x' \text{ värske}}{\to \exists x\ \exists y\ (p(f(x),y) \vee p(y,g(x))) \supset \exists x\ \exists y\ p(x,y)}\,\supset\mathcal{R}$$